

AUTOMATED BONE FRACTURE DETECTION & LOCALIZATION

A 3-Stage Deep Learning Pipeline on FracAtlas

Computer Vision · Final Project Defense

Saint Joseph University · Faculty of Engineering · Spring 2026

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The Clinical Problem

Why bone fracture detection on X-rays matters

178 M

new fracture cases globally each year

(World Health Organization, 2023)

X-ray remains the first-line imaging modality
(low cost, wide availability, fast acquisition)

Why is this hard?

- Hairline cracks, displaced fragments, and subtle cortical disruptions are routinely missed by junior clinicians under emergency time pressure.

Our two-part clinical question

1. Is a fracture present in this X-ray? → classification
2. If so, where exactly on the bone? → localization

Our Approach — A 3-Stage Pipeline

Classify → Localize → Explain

1 Classification

EfficientNet-B3

4-class output:
hand/leg × fractured/normal

86.6% accuracy

2 Localization

YOLOv8s / Faster R-CNN

Bounding boxes around
the suspected fracture

0.62 F1 (FRCNN)

3 Explainability

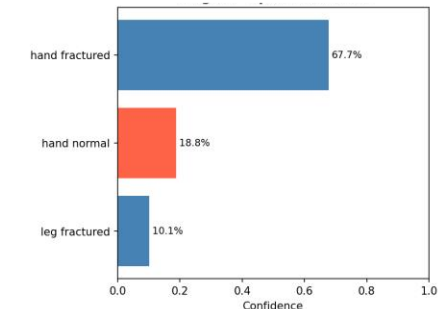
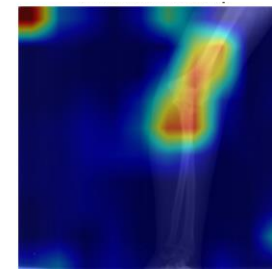
Grad-CAM

Spatial saliency map
over the classifier's input

visual rationale

Key design choice: Stage 2 only runs when Stage 1 flags a fracture

DIAGNOSIS: HAND — FRACTURE DETECTED (68% confidence)



The Dataset Journey

It took us four attempts to find the right corpus

v1

Binary + multi merge

Result:

96% train / 50% val

Why:

Binary dataset (fractured / not fractured) with multi-class one (10 fracture types) for a 2-stage approach.
Failed: mixed modalities, brightness bias ($\mu=51$ vs 84–110)

v2

Same data, fewer classes

Result:

Val ceiling 47–50%

Why:

Merged the 11 classes down to 6.
The val ceiling barely moved: data was the bottleneck, not the model

v3

Single-source switch (pkdarabi)

Result:

Val 32%

Why:

~84 imgs/class for 10 classes — fundamentally insufficient

v4

FracAtlas (Nature)

Result:

Test 86.6% ✓

Why:

Peer-reviewed, single-source, X-ray-only, with bounding-box annotations

Lesson: *no architecture, learning rate, or augmentation strategy can compensate for inconsistent data.*

FracAtlas — Our Final Dataset

Islam et al. (2023), Scientific Data, Nature

4,083 X-rays

From 3 hospitals, annotated by 2 radiologists + 1 orthopedic surgeon

Per-image labels

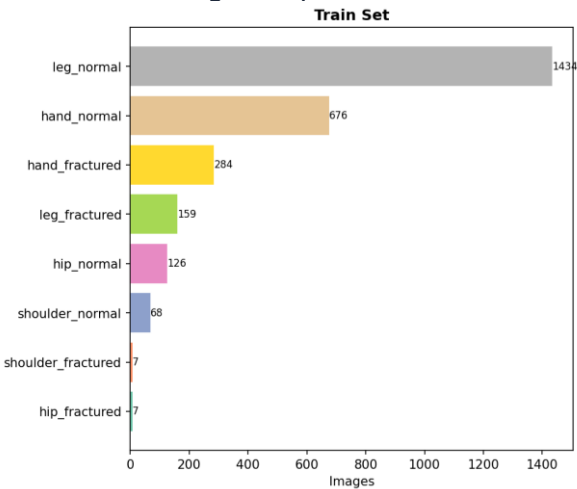
- Body region (hand, leg, hip, shoulder)
- Fracture status (fractured / not)
- Bounding boxes (COCO, VOC, YOLO, VGG)
- Projection view, instance count, hardware



From 8 classes to 4

Class	Train	Status
hand_fractured	284	kept
hand_normal	676	kept
leg_fractured	159	kept
leg_normal	1,434	kept
hip_fractured	7	EXCLUDED
shoulder_fractured	7	EXCLUDED

Why excluded? Only 7 fractured training examples each → 204× imbalance on the 8-class formulation



Stage 1 — EfficientNet-B3 Classifier

Two-phase fine-tuning with class-weighted loss

Why EfficientNet-B3

- 10.7 M parameters
- 300×300 input — preserves the spatial detail of hairline fractures
- ImageNet-pretrained — leverages transfer learning
- Replace 1000-class head with Linear(4) + Dropout(0.4)

No ColorJitter

Brightness jitter would corrupt the diagnostic signal.

Two-Phase Fine-Tuning

Phase 1 (*epochs 1–15*)

Backbone FROZEN • LR_head = $3e-3$

Phase 2 (*epoch 16+, early stop @ patience 12*)

Backbone UNFROZEN • LR_backbone = $1e-5$

LR_head = $1e-4$

Threshold Tuning — Free +7 pp F1

A post-hoc calibration that adds zero training cost

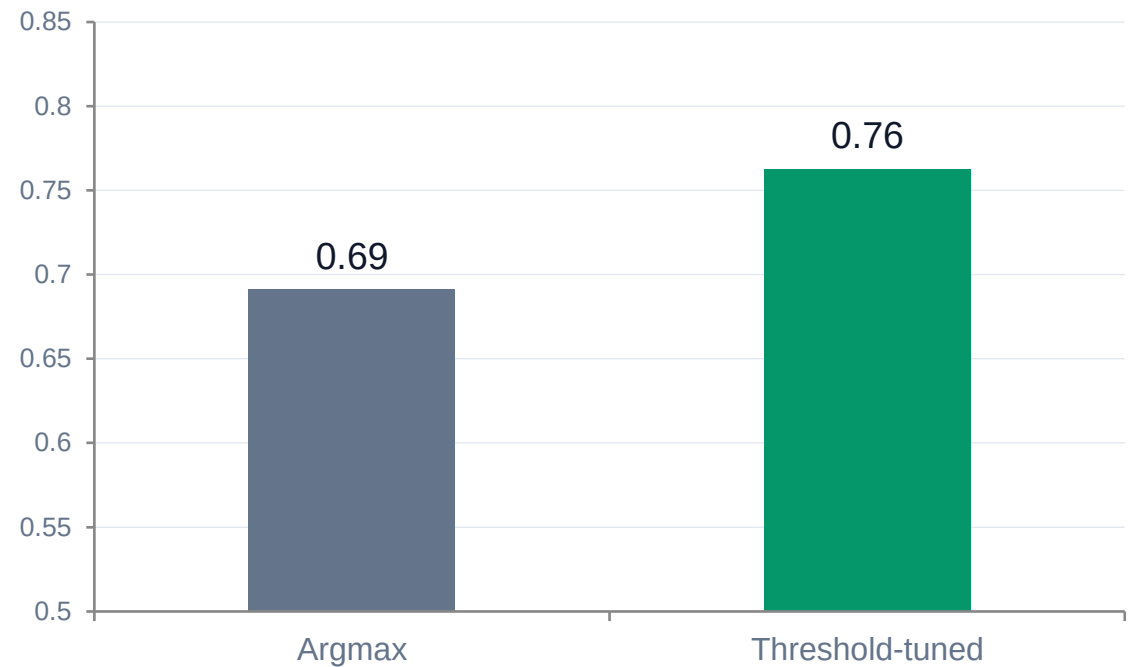
The intuition

- Argmax assumes balanced base rates
- Our classes are imbalanced (9× between leg_normal and leg_fractured)
- Optimal per-class boundary is no longer at $p = 0.5$

Our procedure

Sweep threshold $t \in [0.10, 0.90]$ per class on the validation set; keep the t that maximizes per-class F1. Persist in pkl, reuse at test time.

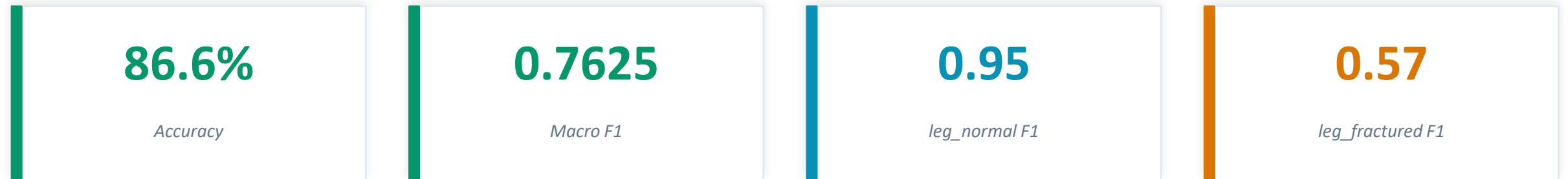
Validation Macro F1 — before vs after



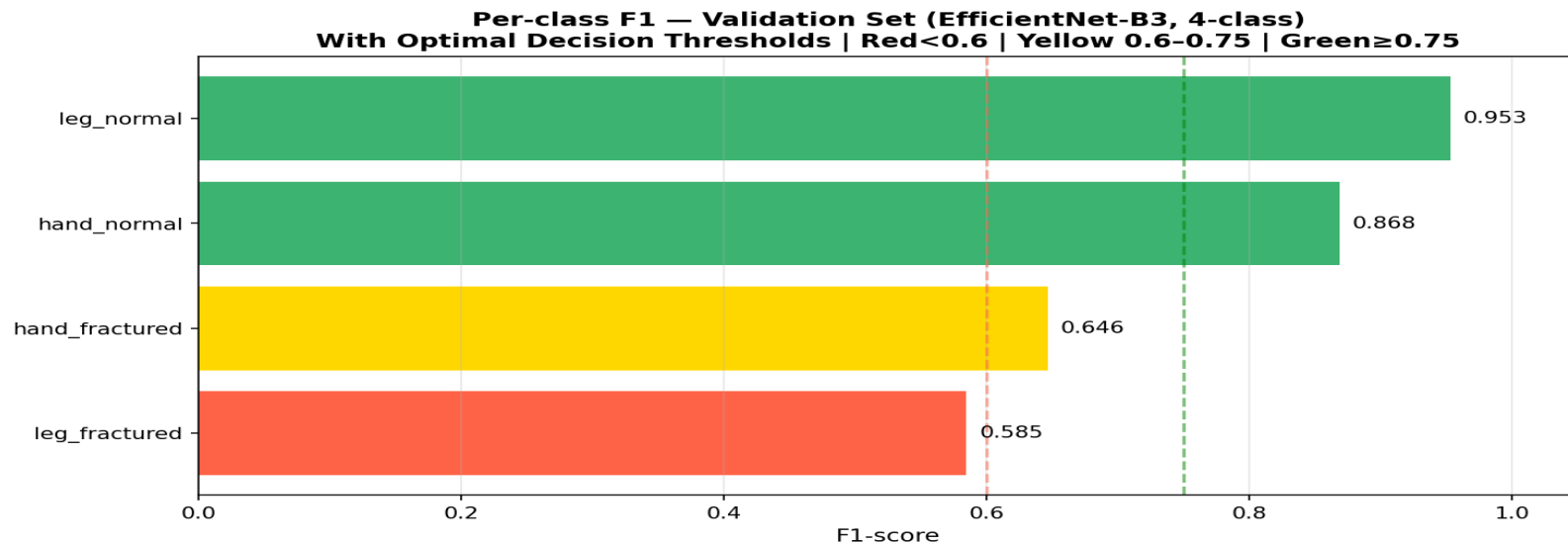
Final thresholds: [0.50, 0.32, 0.50, 0.10]

Stage 1 Results — Test Set (344 images)

Held out from all training and validation



Per-class F1 on the test set



Clinical reading: Recall on leg_fractured (0.59) matters more than precision (0.54) — missing a fracture is costlier than a false alarm.

Stage 2 — Why Two Detectors?

Single-stage vs two-stage: a fair side-by-side comparison

YOLOv8s

1-stage anchor-free detector

- 11.1 M parameters
- 640×640 input
- COCO-pretrained init
- Conservative augmentation (hsv_h = 0 for grayscale)
- No mixup (blurs fracture line orientation)

Optimized for: *speed & precision*

Faster R-CNN

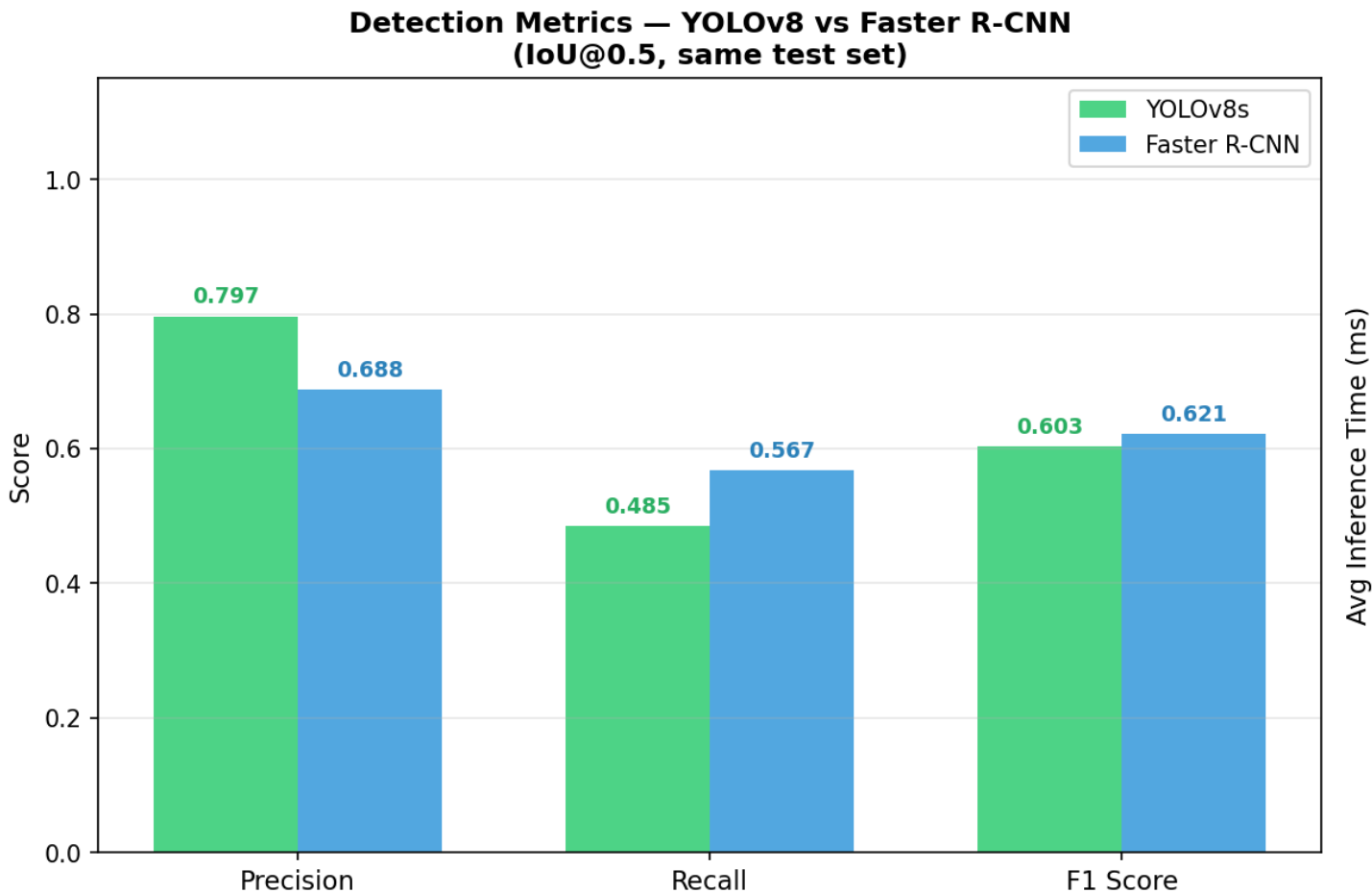
2-stage ResNet-50 FPN detector

- ~41.8 M parameters
- Region Proposal Network → RoI pooling → classification head
- ImageNet-pretrained backbone
- Two-class output (background / fracture)
- Confidence threshold 0.50

Optimized for: *recall & F1*

Detection Comparison — Hand+Leg Filtered Test Set

IoU 0.5 hard matching, identical scope



YOLOv8s wins on...

Precision (+0.08)
Speed: 32 ms / image (4× faster)

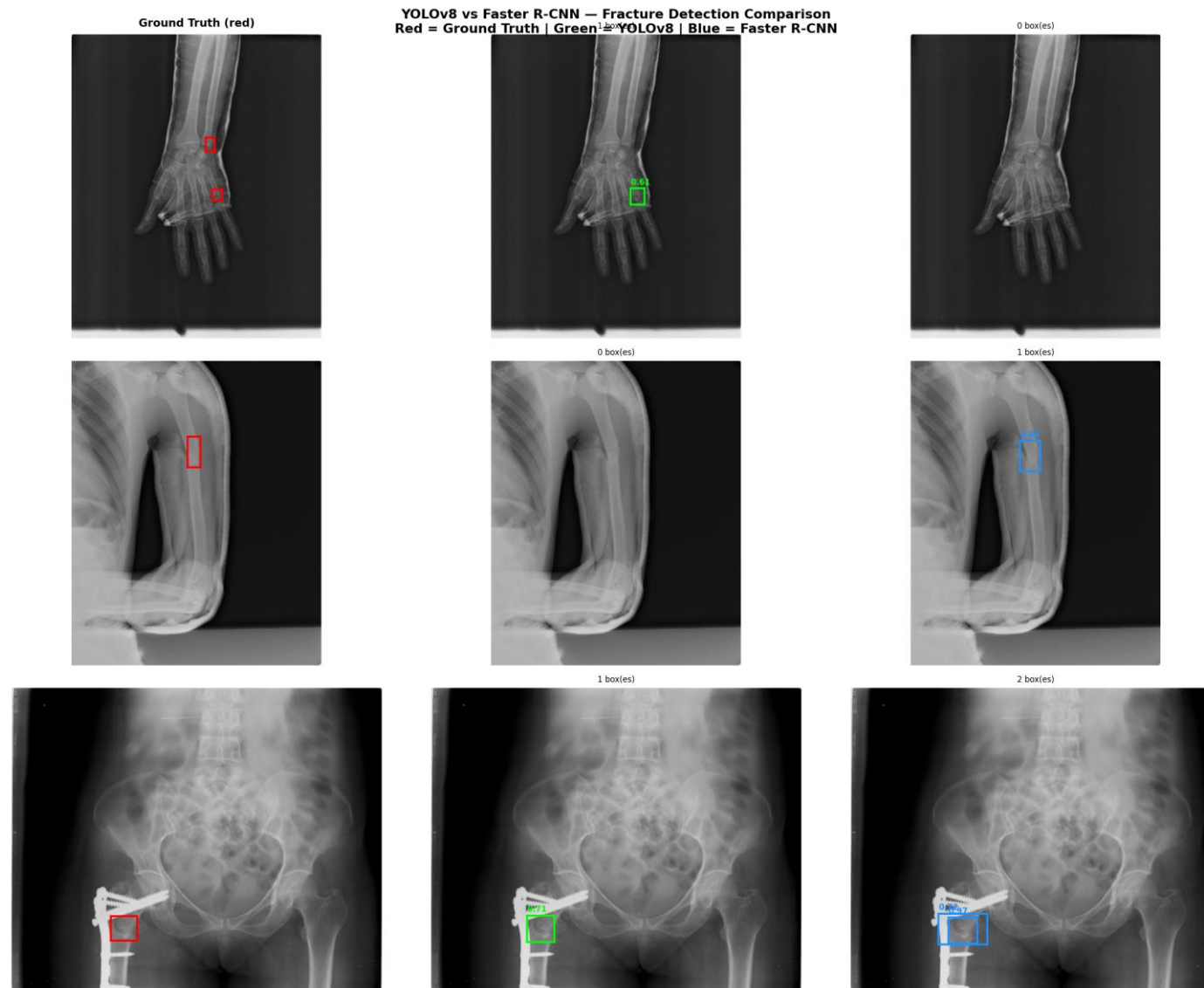
Faster R-CNN wins on...

Recall (+0.09)
F1 (+0.03)
Latency: 126 ms / image

Clinical verdict

Faster R-CNN preferred — missing a fracture is more costly than a false alarm.

Detection Comparison — Hand+Leg Filtered Test Set



Stage 3 — Grad-CAM Explainability

Visual rationale for every classifier decision

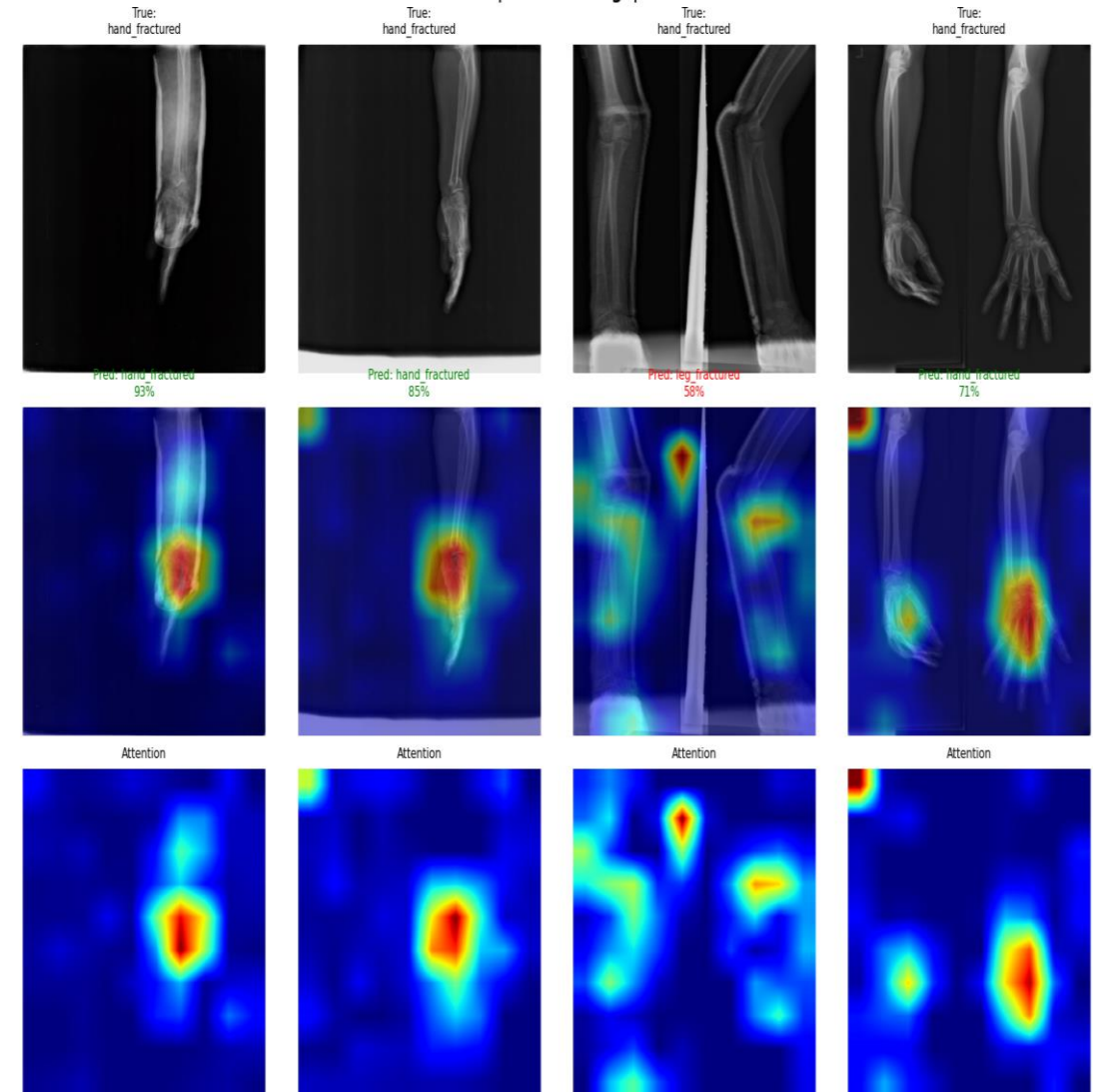
Why explainability matters

- Lets clinicians review the model's reasoning instead of trusting a raw score
- Catches spurious correlations (e.g. background features driving the prediction)
- Required for clinical decision-support tools

Implementation

*Target: features[-1] — the last MBConv block of EfficientNet-B3.
Highest-level features that still preserve spatial resolution.*

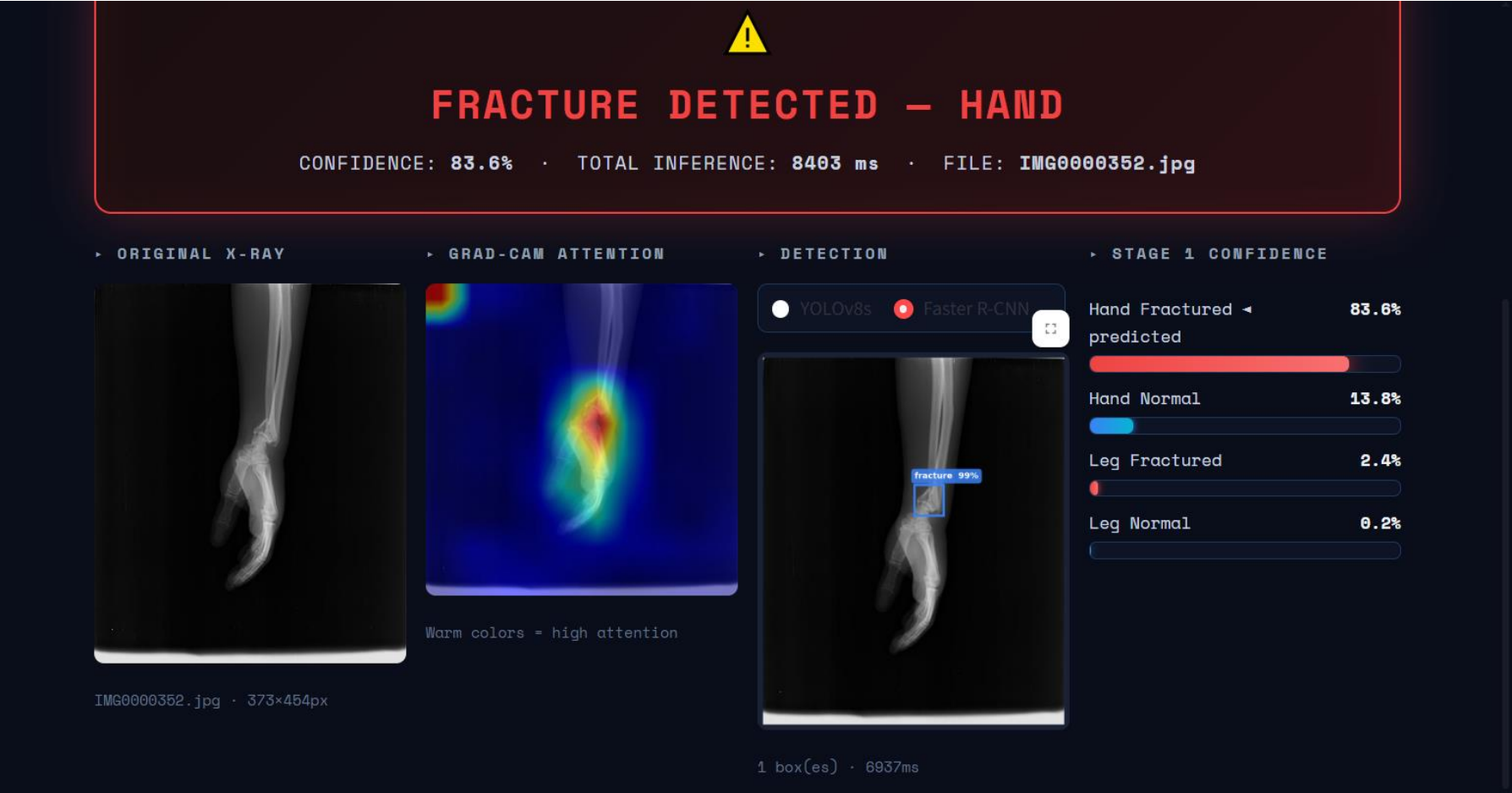
Grad-CAM Visualization — EfficientNet-B3
Row 1: Original | Row 2: Grad-CAM Overlay | Row 3: Heatmap
Green title = correct | Red = wrong | 7/8 correct shown



Warm colors = high attention

Deployed System — Interactive Web Application

Streamlit-based real-time inference on uploaded X-rays



Per-stage panels

Original X-ray, Grad-CAM heatmap, detector bounding boxes, and Stage-1 confidence bars — all in one view.

Detector toggle

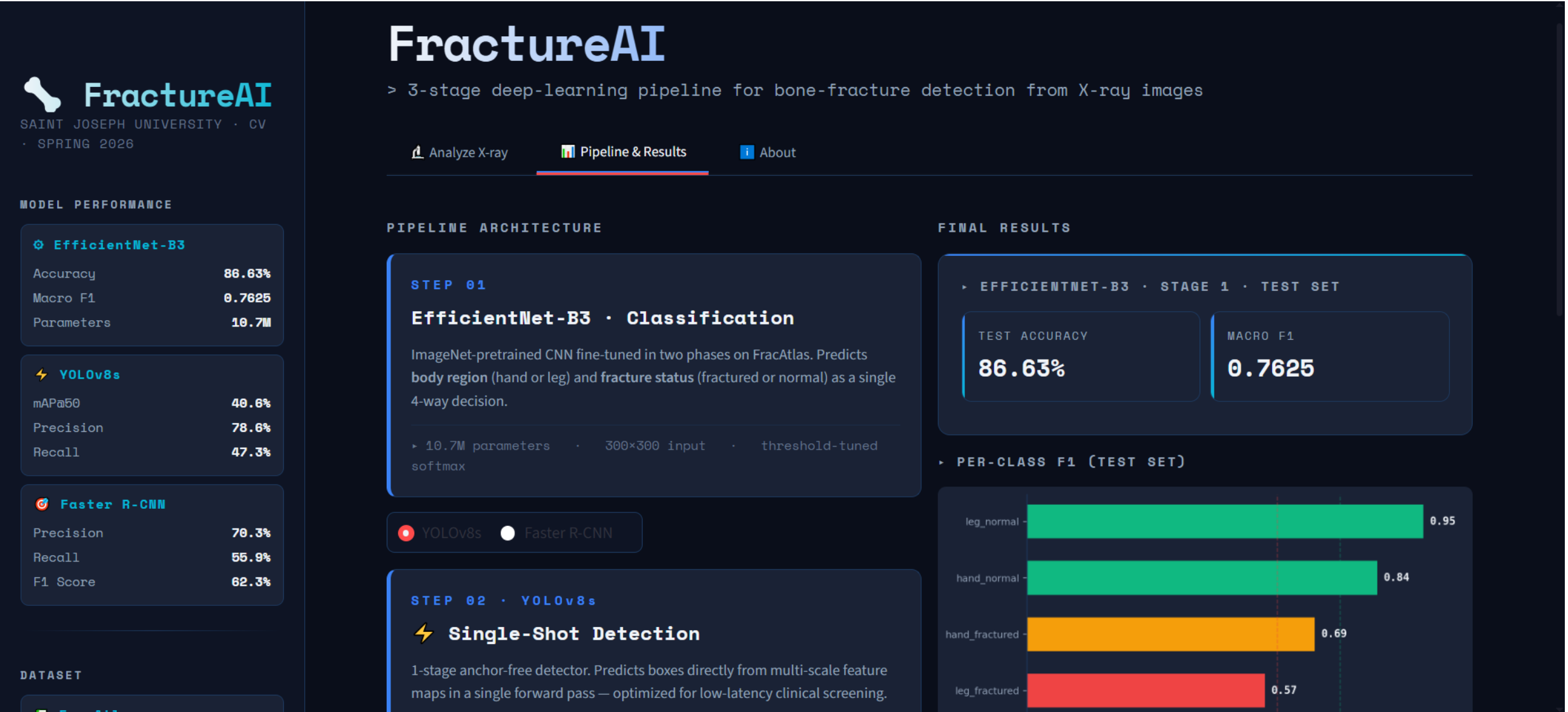
Switch between YOLOv8 and Faster R-CNN at inference time, reusing the cached classifier output.

Calibrated decisions

Stage-1 thresholds [0.50, 0.32, 0.50, 0.10] applied automatically. Detector skipped on normal X-rays.

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Streamlit-based real-time inference on uploaded X-rays



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FractureAI

SAINT JOSEPH UNIVERSITY · CV
· SPRING 2026

MODEL PERFORMANCE

 **EfficientNet-B3**

Accuracy	86.63%
Macro F1	0.7625
Parameters	10.7M

 **YOLOv8s**

mAP@50	40.6%
Precision	78.6%
Recall	47.3%

 **Faster R-CNN**

Precision	70.3%
Recall	55.9%
F1 Score	62.3%

DATASET

Analyze X-ray

Pipeline & Results

About

PROJECT INFORMATION

UNIVERSITY

Saint Joseph University · Beirut

COURSE

Computer Vision · Spring 2026

TASK

3-stage fracture detection pipeline

DATASET

FracAtlas — 4,083 X-rays · 3 hospitals

ANNOTATORS

2 radiologists + orthopedic surgeon

TECHNICAL STACK

Python 3.12

PyTorch 2.x

torchvision

Ultralytics YOLOv8

scikit-learn

pytorch-grad-cam

Google Colab · T4 GPU

Streamlit

matplotlib / seaborn

PIL / NumPy / pandas

MODEL ARCHITECTURE

```
# Stage 1 - EfficientNet-B3 (4-class classifier)
backbone      = efficientnet_b3(weights=ImageNet1k)
classifier     = nn.Sequential(
    nn.Dropout(p=0.4),
    nn.Linear(1536, 4)
)

# Two-phase fine-tuning
phase_1 = freeze_backbone(), lr_head=3e-3, epochs=15
phase_2 = unfreeze_all(),    lr_bb=1e-5, lr_head=1e-4

# Stage 2 - detector ensemble
```

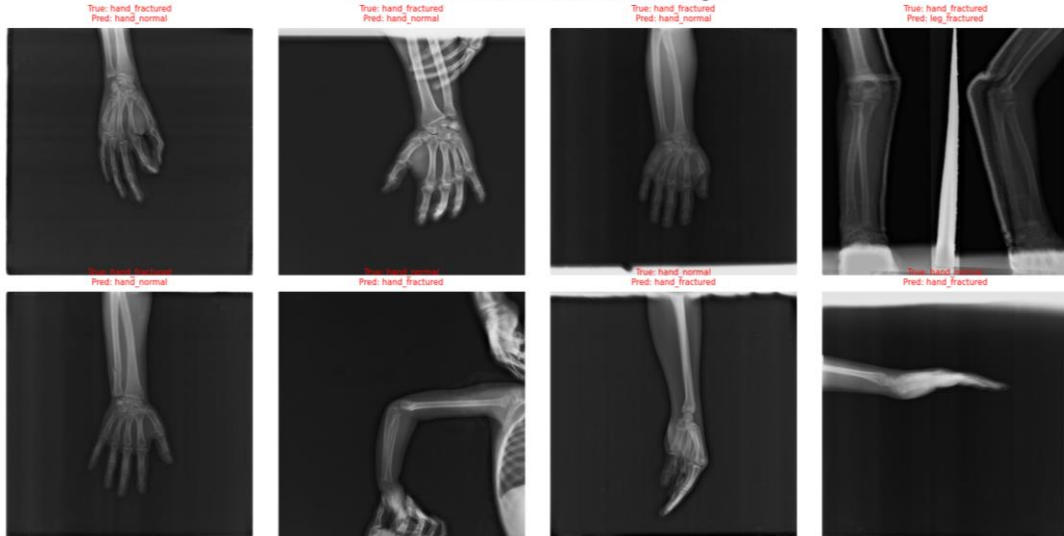
Limitations & Future Work

What we'd do with more time

Honest limitations

- Hand and leg only — hip/shoulder excluded due to scarcity
- Only 3 hospitals — limited inference about scanner robustness
- No external out-of-distribution evaluation

Failure Cases — 100 Misclassified Test Images



Future work (priority order)

1. Hierarchical router: hand/leg vs hip/shoulder so excluded regions are not silently dropped
2. Evaluate on an external dataset to characterize domain transfer
3. Port the pipeline to ONNX / TensorRT for embedded deployment (bonus rubric points)

Conclusion

Three takeaways from this project

1

Dataset quality dominates

Three failed dataset versions taught us that no architecture or LR schedule compensates for inconsistent data. FracAtlas was the turning point.

2

Free wins exist

Post-hoc threshold tuning added ~7 percentage points of Macro F1 with zero training cost. Always calibrate.

3

Fair comparisons matter

Side-by-side YOLOv8 vs Faster R-CNN on a scope-matched test set produced a clinically actionable trade-off — neither model is universally better.

Thank you